

Scalability & Future Plan

Date	02 July 2026
Team ID	SWTID-2026-2030
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority
1	Supports only limited audio formats and file size	Some users may not be able to upload all audio files	High
2	Single-user processing	Multiple users cannot use the system simultaneously	Medium
3	Basic feedback generation	Feedback can be made more detailed and personalized	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports individual users	Deploy on cloud with support for multiple concurrent users
2	Data Storage	Temporary local storage	Integrate cloud database for secure storage of reports and history
3	Performance	Processes one request at a time	Optimize processing and use faster AI models with caching
4	Security	Basic application security	Add user authentication, encrypted data storage, and secure API access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	User Login and Profile Management	1–2 Months	Personalized user experience
Phase 2	Real-Time Voice Recording and Analysis	2–3 Months	Faster and more interactive feedback
Phase 3	AI-Based Personalized Learning Suggestions	3–4 Months	Better concept understanding and learning outcomes
Phase 4	Cloud Deployment with Multi-User Support and Mobile App	4–6 Months	Improved scalability and accessibility across devices